

Coordinated Abrupt Weakening of the Eurasian and North African Monsoons in the 1960s and Links to Extratropical North Atlantic Cooling

Y. LIU

Department of Geography, University of California, Berkeley, Berkeley, California

J. C. H. CHIANG

Department of Geography, and Berkeley Atmospheric Science Center, University of California, Berkeley, Berkeley, California

(Manuscript received 26 April 2011, in final form 29 November 2011)

ABSTRACT

Previous modeling and paleoclimate studies have suggested that cooling originating from the extratropical North Atlantic can abruptly weaken the Eurasian and North African monsoons. The climatic signature includes a widespread cooling over the Eurasian and North African continents and an associated increase to surface pressure. It is explored whether such coordinated changes are similarly exhibited in the observed twentieth-century climate, in particular with the well-documented shift of Sahel rainfall during the 1960s. Surface temperature, sea level pressure, and precipitation changes are analyzed using combined principal component analysis (CPCA). The leading mode exhibits a monotonic shift in the 1960s, and the transition is associated with a relative cooling and pressure increase over the interior Eurasia and North Africa, and rainfall reduction over the Sahel, South Asia, and East Asia. The local circulation changes suggest that the rainfall shift results from the regional response of the summer monsoons to these continental-wide changes. A similar CPCA analysis of atmospheric general circulation model (AGCM) simulations forced by twentieth-century-observed forcings shows similar results, suggesting that origins of the climate shift reside in the sea surface temperature changes, specifically over the extratropical North Atlantic. Finally, an AGCM forced with extratropical North Atlantic cooling appears to simulate these climate impacts, at least qualitatively. The result herein shows that the observed climate signature of the 1960s abrupt shift in Eurasian and North African climate is consistent with the influence of the abrupt high-latitude North Atlantic cooling that occurred in the late 1960s. A definitive causal relationship remains to be shown, and mechanisms elucidated.

1. Introduction

One of the most prominent twentieth-century regional climate shifts was the abrupt shift to drought conditions in the Sahel region of West Africa (Folland et al. 1986; Giannini et al. 2003; Bader and Latif 2003; Held et al. 2005; Rotstayn and Lohmann 2002; Biasutti and Giannini 2006). The Sahel drought started as an abrupt reduction in summer rainfall in the 1960s, and lasted for over two decades with limited recovery toward the present day. Recent work has suggested that, rather than being an isolated local climate shift, the Sahel drought was in fact part of a global climate phenomenon. Baines and

Folland (2007) documented worldwide abrupt shifts in summertime regional climates during the 1960s, particularly in the tropics and over the oceans. They concluded that the Sahel drought was a regional manifestation of a larger near-global climate shift, of which more examples were to be found.

Previous studies, though few, have noted coincident summertime rainfall reductions in other Northern Hemisphere monsoons. Ren et al. (2004) found that the North China portion of the East Asian monsoon region experienced a long-lasting drought since the 1960s, with similar local circulation changes as in the Sahel. Zhang and Delworth (2006) argued that rainfall variability in phase with that over the Sahel also occurred in the South Asian monsoon. Figure 1 compares the observed changes to summer rainfall over the Sahel, North China, and India. All three rainfall regions exhibit similar interdecadal variability, the most striking of which was the

Corresponding author address: Yuwei Liu, 531 McCone Hall, Department of Geography, University of California, Berkeley, Berkeley, CA 94720.
E-mail: yuwei@atmos.berkeley.edu

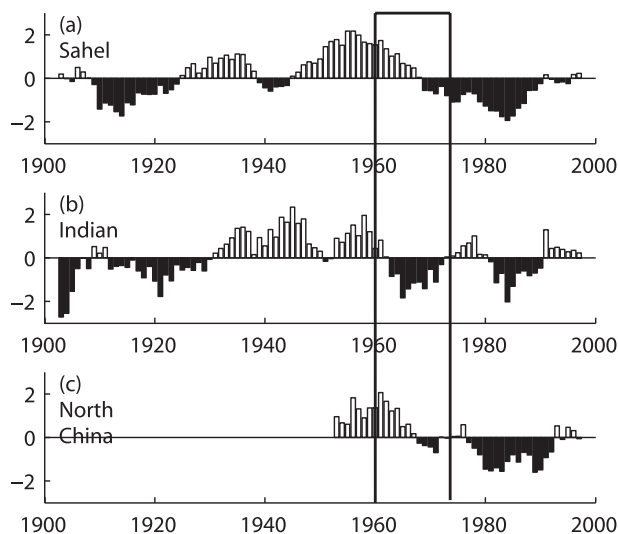


FIG. 1. June–August variation of (a) Sahel rainfall, (b) all-India rainfall, and (c) North China rainfall. Data have been normalized and low-pass filtered with a 7-yr running mean. Source of the indices: Sahel rainfall (Janowiak 1988), North China rainfall averaged over stations in the region of 35° – 45° N, 110° – 125° E [station data from Wang et al. (2000)], and all-India rainfall (from Parthasarathy et al. 1994).

monotonic dip in the 1960s. Whether the coincident changes to these monsoon regions resulted from a common origin, however, remains to be determined.

Monsoon variability in the three regions has been extensively studied, with focus directed toward the influence of tropical sea surface temperature (SST) variability (Giannini et al. 2003; Bader and Latif 2003; Zhou et al. 2008; Kucharski et al. 2009). Giannini et al. (2003) showed that an atmospheric general circulation model (AGCM) forced by global observed SST was able to reproduce the pattern of the Sahel rainfall low-frequency variability, and furthermore attributed the changes to warming in the tropical South Atlantic and the Indian Ocean. Bader and Latif (2003) separated the effect of the two ocean basins and argued that the Indian Ocean warming played a more crucial role. Kucharski et al. (2009) similarly showed that decadal drying of the South Asian monsoon was forced by warming in the Indian Ocean and the Pacific. Zhou et al. (2008) further argued that the tropical Indian and Pacific Ocean warming caused a decreasing trend in global monsoon precipitation in the second half of the twentieth century. They also found the interannual monsoon variability was largely attributed to El Niño–Southern Oscillation (ENSO). These past studies focusing on the role of tropical SST on the monsoons have shed much light on the workings of decadal and interdecadal monsoon variability, in particular for variations during the latter half of the twentieth century.

In this study, we take a different tack by focusing on the monsoon climate shift that occurred during the late 1960s, focusing on the possible role of the extratropical North Atlantic. Our motivation comes from various paleoclimate studies showing that the weakening of the Asian and African monsoons were tied to cooling over the high-latitude North Atlantic (e.g., Stager et al. 2011; Gupta et al. 2003; Shanahan et al. 2009; Wang et al. 2001). The strongest evidence comes from abrupt climate changes during the latter half of the last glacial period ($\sim 50\,000$ to $14\,000$ BP). Greenland ice core records (Dansgaard et al. 1993) show abrupt and large (~ 10 K) transitions between warm “interstadial” states and cold “stadial” states, with the transitions occurring from a few hundred to a few thousand years. The Asian and African monsoons appear to weaken during stadial conditions; for example, a recent study by Stager et al. (2011) shows that the height of Heinrich stadial 1 (16 000–17 000 years ago) coincided with a pronounced megadrought in the Afro–Asian monsoon region. Speleothem records from Hulu Cave in China (Wang et al. 2001) show impressive one-to-one correspondence between stadial events in Greenland to monsoon weakening in Asia throughout the last glacial period. The link between North Atlantic cooling and monsoon weakening also appears to have occurred during the Holocene, albeit at a weaker level; for example, Gupta et al. (2003) show that Holocene weakenings in the South Asian monsoon were tied to North Atlantic cooling, and Shanahan et al. (2009) show similar Holocene linkages, but for the African monsoon.

Prevailing evidence points to the North Atlantic as the origin of these abrupt changes through changes to the Atlantic meridional overturning circulation (AMOC). This idea was first proposed by Broecker et al. (1985), and subsequently supported by many other studies (see Alley 2007). Moreover, coupled model “hosing” simulations that simulate a slowdown of the AMOC through the artificial input of freshwater in the high-latitude North Atlantic (e.g., Vellinga and Wood 2002; Zhang and Delworth 2005; Cheng et al. 2007) show impacts similar to what is inferred from paleodata. An example hosing simulation using the Community Climate System Model, version 3 (CCSM3) is shown in Fig. 2 [the simulation is described in detail in Cheng et al. (2007)]. It shows intense cooling over the extratropical North Atlantic that extends over Eurasia and North Africa (Fig. 2a), and rainfall reduction over the Sahel, India, and North China. A significant weakening of the monsoon winds occur over North Africa, as indicated by the anomalous northeasterly wind (Fig. 2b), coincident with a meridional pressure gradient anomaly between the anomalous high pressure to the north and low pressure

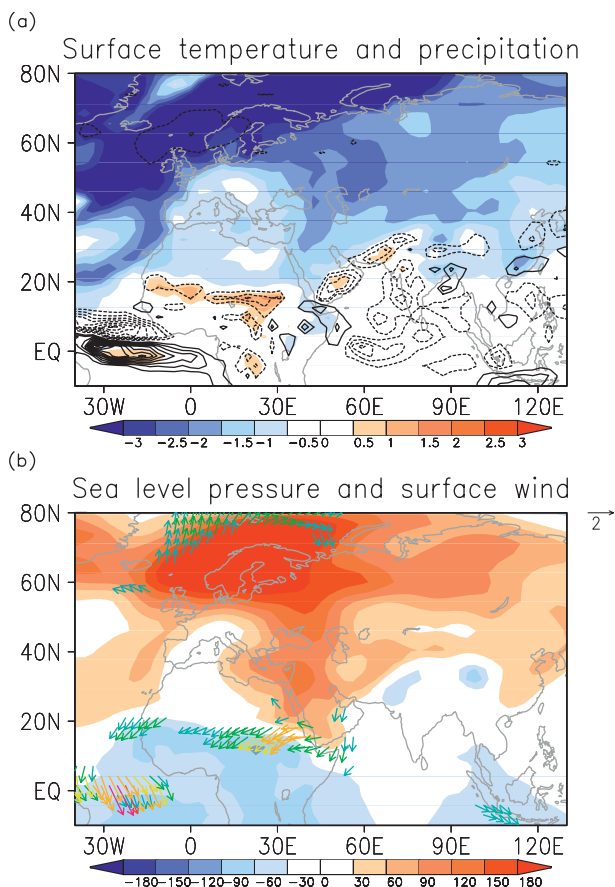


FIG. 2. July–September climate anomalies in years 10–14 of a hosing simulation with the CCSM3, where the upper 970 m of the North Atlantic and Arctic Oceans at 55°–90°N, 90°W–20°E are freshened by an average of 2 psu. Changes in (a) surface temperature (shaded, K) and rainfall (contour interval 0.5 mm day⁻¹, dashed contours are negative) and (b) sea level pressure (shaded, Pa) and surface wind (vectors, m s⁻¹, value under 1 not shown) are shown. Details of this simulation are described in Cheng et al. (2007).

to the south. The model results thus show a continental climate change primarily in surface temperature and sea level pressure (SLP), which may act to bridge the extratropical North Atlantic cooling to the rainfall reduction over the three monsoon systems.

Can this paleoclimate scenario—extratropical North Atlantic cooling leading to a monsoon weakening—be applicable for the twentieth century? In this paper, we advance a hypothesis that this scenario may have occurred, albeit with much-reduced magnitude, during the late 1960s. The seminal paper by Folland et al. (1986) showed that the onset of the Sahel drought was tied to an interhemispheric gradient of Atlantic SST with cooling in the Northern Hemisphere and warming in the Southern Hemisphere (Fig. 3). Apart from the interhemispheric gradient, the striking feature is the intense cooling in high-latitude North Atlantic. This pattern has been

confirmed in subsequent studies, using different statistical techniques and updated global SST and rainfall data. For example, Folland et al. (1991) obtained such a pattern from cross correlation between a Sahel rainfall time series and the Met Office Historical Sea Surface Temperature dataset version 3 (MOHSST3; Bottomley et al. 1990), an updated version of the SST used in Folland et al. (1986). They also captured this pattern using an empirical orthogonal functions (EOF) analysis of the SST data. Rowell et al. (1995) obtained similar results using MOHSST4 and a gridded rainfall dataset (Hulme 1992). Recently, Thompson et al. (2010) showed that the interhemispheric gradient suffered an abrupt shift in the late 1960s, and in particular showed sizable abrupt cooling in the high-latitude North Atlantic SST. Thus, the parallel to the abrupt climate change during the last glacial–extratropical North Atlantic cooling and weaker monsoon appears to occur during the late 1960s shift.

Multidecadal changes to basinwide North Atlantic SST is commonly attributed to the Atlantic multidecadal oscillation (AMO), a low-frequency oscillation of the entire (as opposed to high latitude) North Atlantic basin; the AMO also shows a cooling trend throughout the 1960s and 1970s.¹ Paralleling the paleoclimate associations, the AMO has been postulated to be the result of variations to the Atlantic meridional overturning circulation (e.g., Knight et al. 2005). Furthermore, the AMO has also been linked to rainfall reduction in India and the Sahel (Zhang and Delworth 2006; Kucharski et al. 2009).

Several other studies have hinted that the pattern of climate changes occurring with the 1960s monsoonal weakening may resemble those associated with extratropical North Atlantic cooling. Haarsma et al. (2005) found that Sahel rainfall was related to mean SLP over the Sahara, north of the Sahel, which in turn was related to surface temperatures over Eurasia and North Africa. Biasutti et al. (2009) examined lead–lag correlation between the Sahara SLP and the Sahel rainfall, and argued that variation in the former caused that in latter, rather than vice versa. For the Asian monsoon, D’Arrigo et al. (2006) found from proxy data that its magnitude was correlated to Eurasian surface temperature, while Goswami et al. (2006) showed its onset and withdrawal depends on tropospheric temperature over Eurasia for modern periods. Taken together, these studies raise the possibility that the 1960s’ shift in the Eurasian and North African monsoons were tied together through temperature (and hence SLP) changes over the interior Eurasia and North Africa, perhaps caused by extratropical North Atlantic

¹ We discuss the relationship between the AMO and the abrupt cooling of the high-latitude North Atlantic in section 7.

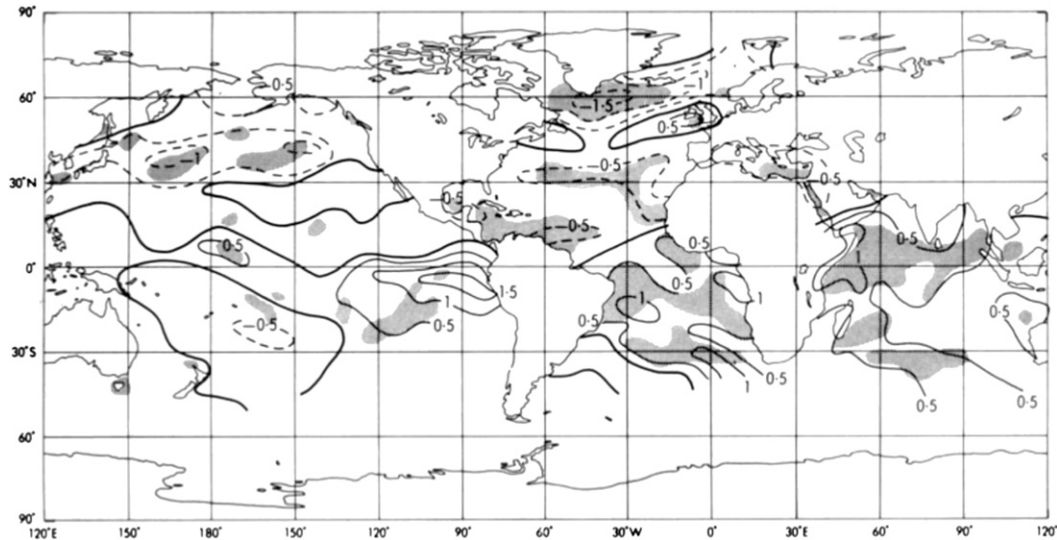


FIG. 3. From Folland et al. (1986), global boreal summer SST anomalies associated with the drought period over the Sahel. Plotted is SST: the July–September average of 1972–73 and 1982–84 (Sahel dry) minus the average of 1950, 1952–54, and 1958 (Sahel wet). The contour interval is 0.5°C , and shaded are regions where the difference is significant (at the 90% level, according to a t test). Reproduced with permission from *Nature*.

cooling. We speculate that the monsoonal rainfall weakening was the regional manifestation of this interior continental temperature and SLP shift, leading to a decreased meridional pressure gradient in the monsoon regions, which ultimately weakened the monsoonal rainfall.

This study explores the possibility of a coordinated weakening of the Sahel and Eurasian summer monsoons in the 1960s from cooling over the Eurasian and North African continental interiors from both observational analysis and analysis of twentieth-century simulations. In the next section, we briefly introduce the datasets and describe the methods. In section 3 we present features of the observed Eurasian and North African climate shift, followed by the response of regional monsoons in section 4. We turn to twentieth-century climate simulations to explore the cause of the shift in section 5, and further examine the cause identified with an atmospheric general circulation model simulation in section 6. The results are summarized and discussed in section 7.

2. Data and methods

a. Data

Our hypothesis involves changes to surface temperature and SLP over the continental interior linked to rainfall in the monsoon regions, so we use those three fields for our data analysis. We combine reanalysis and observational data into two distinct sets. The first set of data consists of surface temperature and SLP from National Centers for Environmental Prediction–National Center for Atmospheric Research (NCEP–NCAR) reanalysis

(Kalnay et al. 1996) and the instrumental precipitation anomaly compiled by Dai et al. (1997) from the National Aeronautics and Space Administration (NASA) Goddard Institute for Space Studies (GISS), both on a $2.5^{\circ} \times 2.5^{\circ}$ grid (NCEP–DAI). We avoid using reanalysis precipitation because rainfall is not assimilated and therefore is not a modeled quantity. The second set of data is solely from observations, that is, $5^{\circ} \times 5^{\circ}$ Northern Hemisphere SLP data from NCAR (Trenberth and Paolino 1980), and $2.5^{\circ} \times 3.75^{\circ}$ precipitation and $5^{\circ} \times 5^{\circ}$ surface temperature anomalies compiled by the Climatic Research Unit (CRU) of the University of East Anglia (Brohan et al. 2006; Hulme 1992; NCAR–CRU). We will show that we obtain similar results from these two different sets, indicating that our results are robust.

We undertake a similar analysis using output from twentieth-century simulations of the Atmospheric Model Intercomparison Project (AMIP). The first set of runs is forced by global observed time-varying SSTs from January 1950 through December 2000 plus volcanic, greenhouse gas, aerosol, and solar forcings, as used in the Intergovernmental Panel on Climate Change (IPCC) twentieth-century experiments (20C3M), and is referred to as IAMIP runs. The SST dataset used by the simulation was produced by merging the monthly mean Hadley Centre sea ice and SST dataset and the National Oceanic and Atmospheric Administration weekly optimum interpolation SST analysis (Rayner et al. 2003; GFDL Global Atmospheric Model Development Team 2004; Hurrell et al. 2008). IAMIP runs from two AGCMs are used—one is the NCAR Community Atmosphere

Model, version 3 (CAM3; Collins et al. 2006) and the other is the Geophysical Fluid Dynamics Laboratory (GFDL) Atmosphere Model, version 2.1 [AM2.1 (Delworth et al. 2006)]. The CAM3 uses a T42 grid and has 17 vertical levels, and the AM2.1 is a 2.0° latitude $\times$ 2.5° longitude grid and has 24 vertical levels. We also analyze a second set of simulations [vanilla AMIP (VAMIP)], which has no external forcings beyond the global time-varying SST; only the CAM3 simulations were available in this format.

b. Methods

We make use of combined principal component (PC) analysis (CPCA) in our study in order to extract the leading mode of joint variability of all three fields. The PCA, also referred as EOF analysis, is commonly used to identify dominant modes (PCs) of variability of single climate fields and separate them with different spatial structures (Wilks 2006). For the main analysis, PCA is applied to the combination of the three fields (surface temperature, SLP, and precipitation), which have been normalized prior to applying the PCA. Because the data we use have fewer time points than grid points, the CPCA extracts coupled patterns more accurately than other statistical methods, such as singular value decomposition or single-field-based PCA (Bretherton et al. 1992).

All fields are averaged over June–August to obtain the summertime data. Prior to analysis, the global mean temperature is subtracted from the temperature field. This was done because we are interested not in temperature over Eurasia and North Africa per se, but rather its temperature relative to the global surroundings. Subtracting the global mean temperature brings out the spatial gradient of the temperature field in our analysis.

We choose a domain of analysis consistent with our hypothesis that the weakening monsoons over Eurasia and North Africa are tied to climate changes over the continental interior. It incorporates the Eurasian and North African landmasses: from 2.0° to 77.0°N , and from 17°W to 132°E , and we use data only over land regions. The time period of our analysis is 1950–98, and only grid points with more than 30 yr of data available are used. All data are smoothed with a 7-yr running mean so that variability over decadal scales is highlighted. We normalize the data at each grid point prior to analysis, and thus use the correlation matrix for the PCA.

3. Observational analysis of the large-scale climate shift

a. A preview

As a first cut in showing the 1960s regime shift, we show the composite difference for the summertime (June–August)

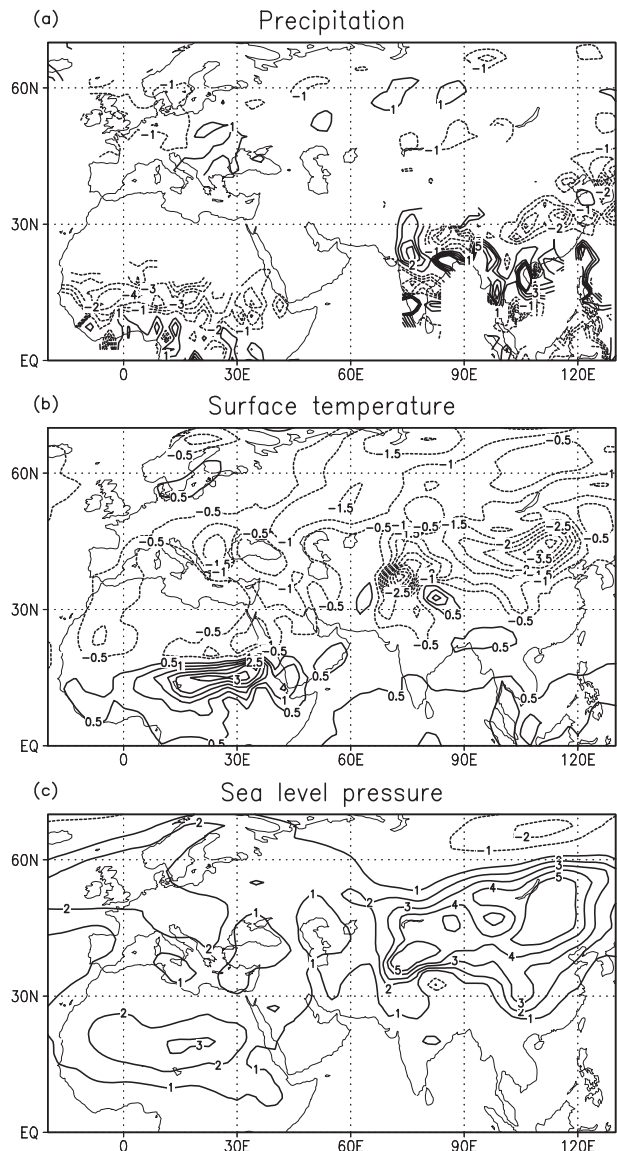


FIG. 4. June–August composite for the period of 1970–79 relative to the period of 1950–59 from NCEP–DAI for (a) precipitation (cm month^{-1}), (b) surface temperature (K), and (c) SLP (hPa). Dashed contours are negative.

months between the 1970–79 and 1950–59 decades (Fig. 4), using the NCEP–DAI dataset. We use data only up to 1979 to avoid potential artifacts within reanalysis data resulting from the introduction of satellite observations in the assimilation after 1979. The anomaly fields appear to capture the climate changes that we hypothesize. There is a reduction in rainfall across the Sahel and North China; the Indian rainfall change shows alternative drying and wetting from the northeast to southwest, but there is an overall reduction. There is cooling over the continental interiors with maxima over East Asia and

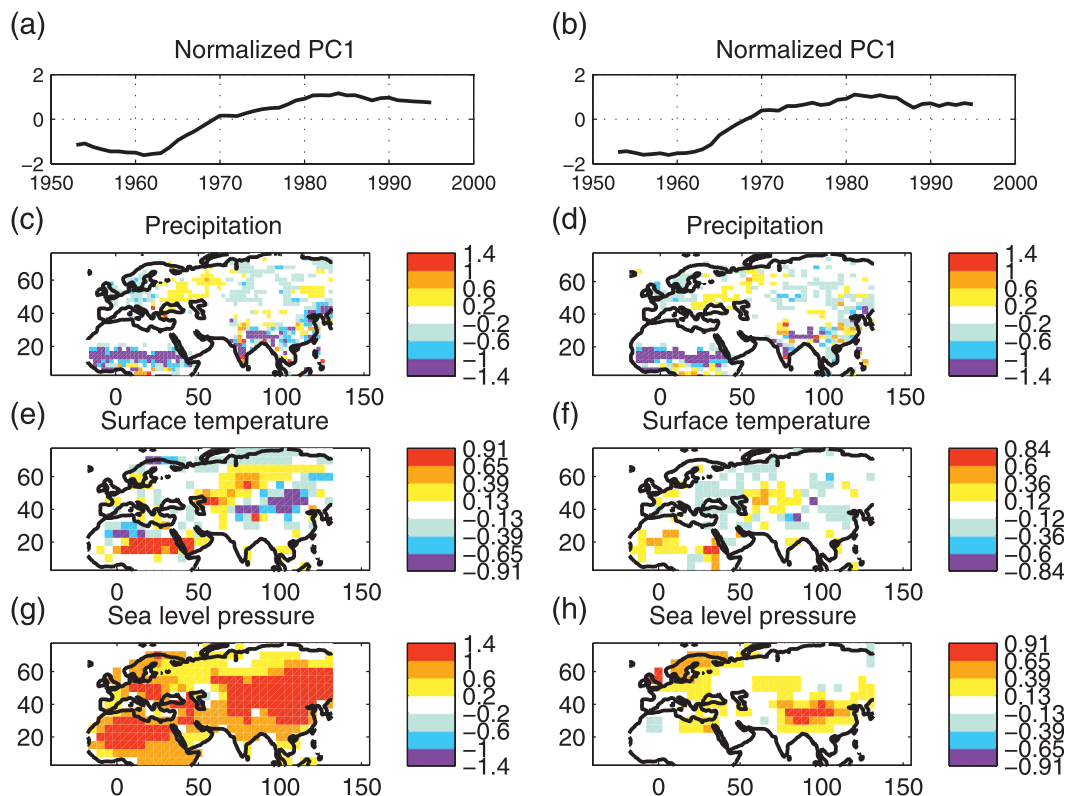


FIG. 5. Leading EOF from CPCA for (left) NCEP-DAI and (right) NCAR-CRU. (a),(b) Normalized PC1. Spatial pattern is shown as regressions of (c),(d) precipitation, (e),(f) surface temperature, and (g),(h) SLP onto the PCs.

eastern central Asia, and there is a near-uniform SLP increase over Eurasia and North Africa, with maxima over East Asia and eastern central Asia, and two local maxima over Europe and North Africa.

b. CPCA analysis

The leading mode from the CPCA of the summertime (June–August) NCEP-DAI data is shown in the left panel of Fig. 5. This mode exhibits the 1960s' climate shift, explaining 32% of the variance, compared with the second EOF that explains 17%. The normalized PC1 (Fig. 5a) indicates that the three fields started changing monotonically at the beginning of the 1960s, and most of the change is achieved by the end of the decade. This new state is essentially maintained throughout the rest of the analysis period into the 1990s. The associated spatial patterns (calculated as the regression onto PC1) resemble the composites shown in Fig. 4. The rainfall weakening is evident in all three monsoon regions (Fig. 5c). The leading mode extracts the Sahel drought as a band of negative values between 5° and 20°N that extends across the entire latitude band within Africa. The east China “north drought–south flooding” rainfall

pattern is similarly extracted, characterized by a dipole structure with a negative anomaly in North China and positive anomaly in the Yangtze River valley (Xu 2001; Li et al. 2010). The rainfall changes over India are more complex, but have drying over most of the Indian Subcontinent, except for its northern extremity. The temperature structure (Fig. 5e) shows alternative cooling and warming in the midlatitudes over Eurasia and North Africa. East Asia and eastern central Asia cools, as does North Africa north of the Sahel. On the other hand, there is surface warming located directly over the Sahel, but this is consistent with the rainfall reduction there. The major difference from composite analysis (Fig. 4) is the warming over central Asia. With the cooling upstream and downstream, it appears to form a stationary wave pattern over midlatitude Eurasia. The pressure pattern (Fig. 5g), however, shows a uniform increase with maxima coinciding with regions of maximum cooling.

PCs derived from CPCA may be dominated by a single field with largest variance, and thus may not be representative of joint variability in all the fields (Bretherton et al. 1992). To address this, we applied PCA to each individual field used in the NCEP-DAI analysis

(figures are not shown). The leading PCs of both surface temperature and SLP resemble those from the CPCA, as does the spatial pattern. Thus, in terms of these fields, the 1960s shift appears to be robust. The shift appears to be somewhat less robust if the individual PC analysis is applied to precipitation only. Instead of a shift in the 1960s, the leading EOF of the Dai et al. (1997) precipitation expresses a decreasing trend from 1960 to 1985 over the monsoon regions of interest. The somewhat ambiguous identification of the 1960s' shift may be a result of mixing signals with Asian rainfall reductions in the 1970s in response to the Pacific decadal oscillation (PDO; Li et al. 2010), but it may also be due to the inherent nature of precipitation data. Precipitation is spatially and temporally noisy, and signals are not easily extracted; furthermore, the precipitation dataset suffers from incomplete data. As noted by Dai et al. (1997), a number of Northern Hemisphere land stations available for their precipitation compilation decreased by over 20% since the late 1970s, making them less reliable after that time. For some regions over Asia with the maximum inconsistency, station density decreased to as low as one station per grid (Fig. 1 in Dai et al. 1997), making the sampling error as large as 45%. For comparison, we applied PCA to the precipitation data from CRU (figures not shown). Unlike the PC using the NCEP–DAI rainfall dataset, the leading PC of the NCAR–CRU resembled that of NCEP–DAI's CPCA, as did the spatial pattern of monsoon rainfall changes. We thus conclude that while the results of the PCA applied to precipitation are mixed, there are good reasons for this having to do with the uncertainty in the precipitation datasets. Overall, the individual PCA analyses support the robustness of our combined PCA analysis.

The same CPCA analysis using the NCAR–CRU observational dataset, which consists of SLP from NCAR and surface temperature and precipitation from CRU, also demonstrates the 1960s' shift in CPCA (Fig. 5, right panel). The shift is captured in the leading EOF, with 23% of the variance explained. Both the PC and spatial structure are nearly identical to those of the NCEP–DAI analysis. The PC captured the jump during the 1960s to a new level after the 1970s (Fig. 5b). The spatial patterns of temperature and SLP anomalies from NCAR–CRU, although of a smaller magnitude, retain the main features in those from NCEP–DAI analysis, including cooling over Eurasia with a maximum over eastern central Asia, slight warming over central Asia and the Sahel, and the near-uniform increase in SLP (Figs. 5f, h). Both the extent and magnitude of rainfall anomalies resemble those from NCEP–DAI analysis (Figs. 5c,d), as well as its individual PCAs. This analysis, using

purely observational (as opposed to reanalysis) datasets, demonstrates that the result of the previous CPCA analysis is not an artifact of using the NCEP reanalysis.

4. Response of monsoon circulation

In this section, we characterize the regional circulation changes associated with the 1960s' shift in the monsoon regions of interest.

a. Sahel

The circulation changes accompanying Sahel drought have been extensively studied (e.g., Grist and Nicholson 2001; Nicholson 2009), and we draw connections from them to our results. Cooling over Eurasia and North Africa results in a shallower Sahara thermal low, weakening the land–sea pressure gradient and thus westerlies transporting moisture from the tropical ocean (Haarsma et al. 2005; Biasutti et al. 2009). Nicholson (2009) showed that the strength of the westerlies was strongly correlated with the surface pressure gradient, and suggested that weaker westerlies led to the Sahel rainfall reduction. Figure 6 shows the regional climate over the Sahel during 1950–59 and its response to the 1960s' climate shift. Near the surface, the SLP increase and weakening westerlies are apparent, as is the rainfall reduction over the Sahel (Figs. 6d,f). In the low and middle troposphere, both moist static energy (MSE) and moisture convergence decrease across the Sahel latitude bands, especially in the middle and east, while moisture convergence strengthens along the southern coast. The MSE is analogous to equivalent potential temperature, and (under the assumption of a fixed upper-tropospheric MSE) the lower-tropospheric MSE is a measure of the moist convective instability. The decrease in lower-troposphere MSE indicates a more stable atmosphere over the Sahel. Moreover, the dipole in the moisture convergence anomaly indicates a southward displacement of land ITCZ. Cooling extends throughout the midtroposphere, and the consequent decrease in the meridional gradient of mean tropospheric temperature weakens the upper-level tropical easterly jet (Fig. 6b), similar to what was found in Nicholson (2009). Nicholson's analysis suggested that the change in strength of this jet influences convection by modulating upper-level divergence; a weak jet led to rainfall reduction over the Sahel.

b. East Asia

Li et al. (2010) previously examined a climate shift over the East Asian summer monsoon region associated with changes to the PDO, using the NCEP–NCAR reanalysis and twentieth-century climate runs from CAM3

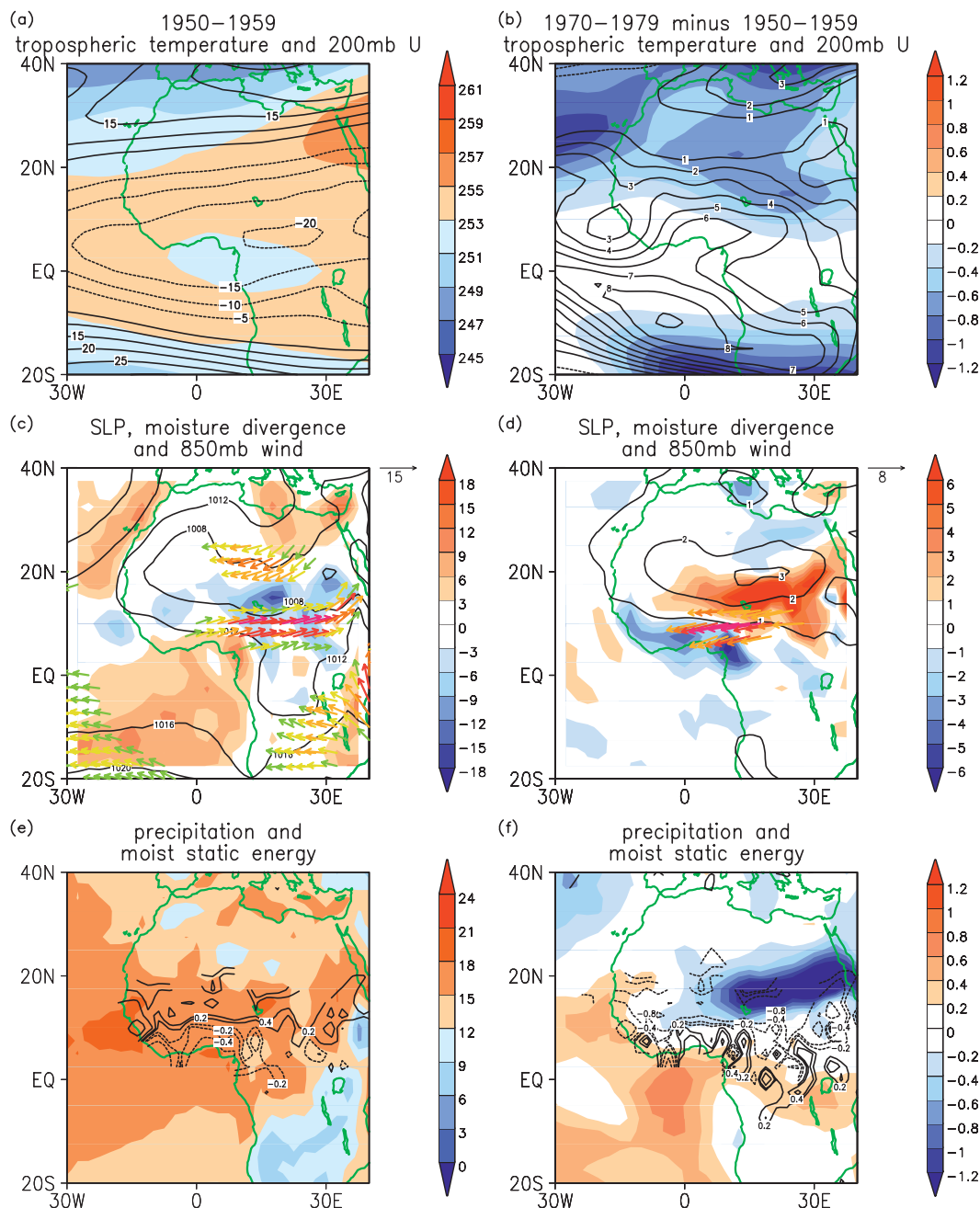


FIG. 6. June–August climate change over the Sahel. (a),(b) Mean tropospheric temperature (shaded, K) and 200-mb zonal wind (contours, m s^{-1} , dashed contours are negative). (c),(d) Mass-weighted integral of divergence of moisture transport from surface to 500 mb (shaded, $10^{-5} \text{ kg s}^{-1} \text{ m}^{-2}$), sea level pressure (contours, hPa), and 850-mb wind (vectors, m s^{-1} , value under 6 not shown). (e),(f) Mass-weighted integral of moist static energy from surface to 700 mb (shaded, 10^7 J m^{-2}) and precipitation anomaly (contours, mm day^{-1} , the blank spots indicate missing data). (a),(c),(e) 1950–59 mean and (b),(d),(f) 1970–79 minus 1950–59 difference. Mean tropospheric temperature is defined as temperature mass-weighted averaged from 600 to 200 mb. The precipitation anomaly is with respect to the 1951–79 period. All data are from NCEP reanalysis, except the precipitation is from GISS/DAI precipitation.

and GFDL AM2.1 (similar model simulations are used in our study). They identified a southern flooding–northern drought rainfall pattern, accompanied by a weakening of the 850-hPa southwesterly winds and southward shift of the 200-hPa jet stream. They attributed these changes to tropical SST forcings.

Following Li et al. (2010), we reexamined the monsoon index they employed, the East Asian Summer Monsoon Index (EASMI), which is defined as the normalized zonal wind shear between 850 and 200 hPa. In our case (and different from the Li et al. analysis), we computed this quantity only over 20°–40°N, 110°–120°E land regions. The index generally represents the strength of baroclinicity over the land monsoon region. The index (Fig. 7) demonstrates a decreasing trend, as stated in Li et al. (2010). We employed a regime shift detection analysis on this time series using the sequential data-processing technique introduced by Rodionov (2004); the result shows a shift occurring in the 1960s, indicating an abrupt weakening of the East Asian monsoon during that period, in accordance with our hypothesis.

The regional climate changes associated with this shift are shown in Fig. 8. In the 1950s' climatology, westerlies prevail in the upper troposphere over East Asia (Fig. 8a). Near the exit of the upper westerly jet, a surface low pressure region exists, which acts to drive southerly winds toward eastern China from the South China Sea (Fig. 8c). The summer precipitation mainly falls over the central and northern parts of eastern China (Fig. 8e). During the 1960s' climate shift, the mean tropospheric temperature cooled by 2°C over North China (Fig. 8b), and the upper-level westerlies strengthened on its southern edge (Fig. 8d). This strengthening reduced the northward progression of the East Asian monsoon, resulting in drought in North China and excessive rain to the south (Yu and Zhou 2007). There are also SLP increases over northeastern China and a consistent low-level northerly wind anomaly, indicating a weakening of the East Asian monsoon (Fig. 8d). Reduction in the lower-tropospheric MSE is found over North China, similar to the Sahel. The anomalous moisture divergence (convergence) occurring over eastern (southern) China is consistent with the rainfall changes. The moisture convergence anomaly over North China runs counter to the observed rainfall reduction; however, an examination of the surface latent heat flux suggests that the convergence is largely offset by decreased evaporation (figures not shown).

c. South Asia

The climate anomalies associated with the 1960s' shift over South Asia are comparable in strength to the other monsoon regions, but are spatially complex. As in East

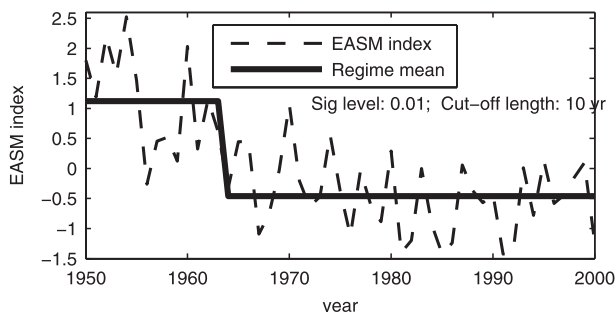


FIG. 7. Normalized EASMI index (dashed line) and its regime mean (solid line) after applying a regime shift detection analysis. The regime mean shows two distinct states before and after 1967. The significant level is 0.01 and cutoff length is 10 yr for the regime detection.

Asia, tropospheric cooling and SLP increase are present north of the summer monsoon regions (Fig. 8b). The rainfall decreases over northeastern and central India, and increases at the southern flanks of these two regions (Figs. 8d,f); these rainfall changes suggest a southward displacement of the monsoon rainfall belt toward the ocean in response to the cooling in the north. Goswami et al. (2006) analyzed tropospheric temperature changes between 1950–60 and 1970–80 (similar to the periods chosen in our analysis) and suggested that the cooling could also lead to an earlier withdrawal of the monsoon rainy season.

5. The climate shift in twentieth-century simulations

In this section, we present results of CPCA applied to model simulations of the twentieth century to see if the climate shift is simulated and, if so, to infer causes.

The ensemble means of the twentieth-century forced SST IAMIP run of GFDL AM2.1 (10 members) and CAM3 (5 members) are first analyzed. Both model simulations captured the observed climate shift as the leading mode, with 45% (GFDL AM2.1) and 35% (CAM3) variance explained, respectively (Fig. 9). The respective PCs resemble those derived from the observational analysis, with a similar shift occurring in the 1960s. The GFDL model was able to simulate the observed Sahel drought and north drought–south flooding pattern in China. For India, a southward movement of the rainbelt is suggested from the pattern of drying in the north and increased rainfall in the south, which is qualitatively consistent with the observations while differing in details (Fig. 9b). The temperature changes show the correct relative strength over Eurasia, including the cooling maxima over east and eastern central Asia, and less cooling in central Asia, where some warming is seen in observations (Fig. 9e). Significant surface warming

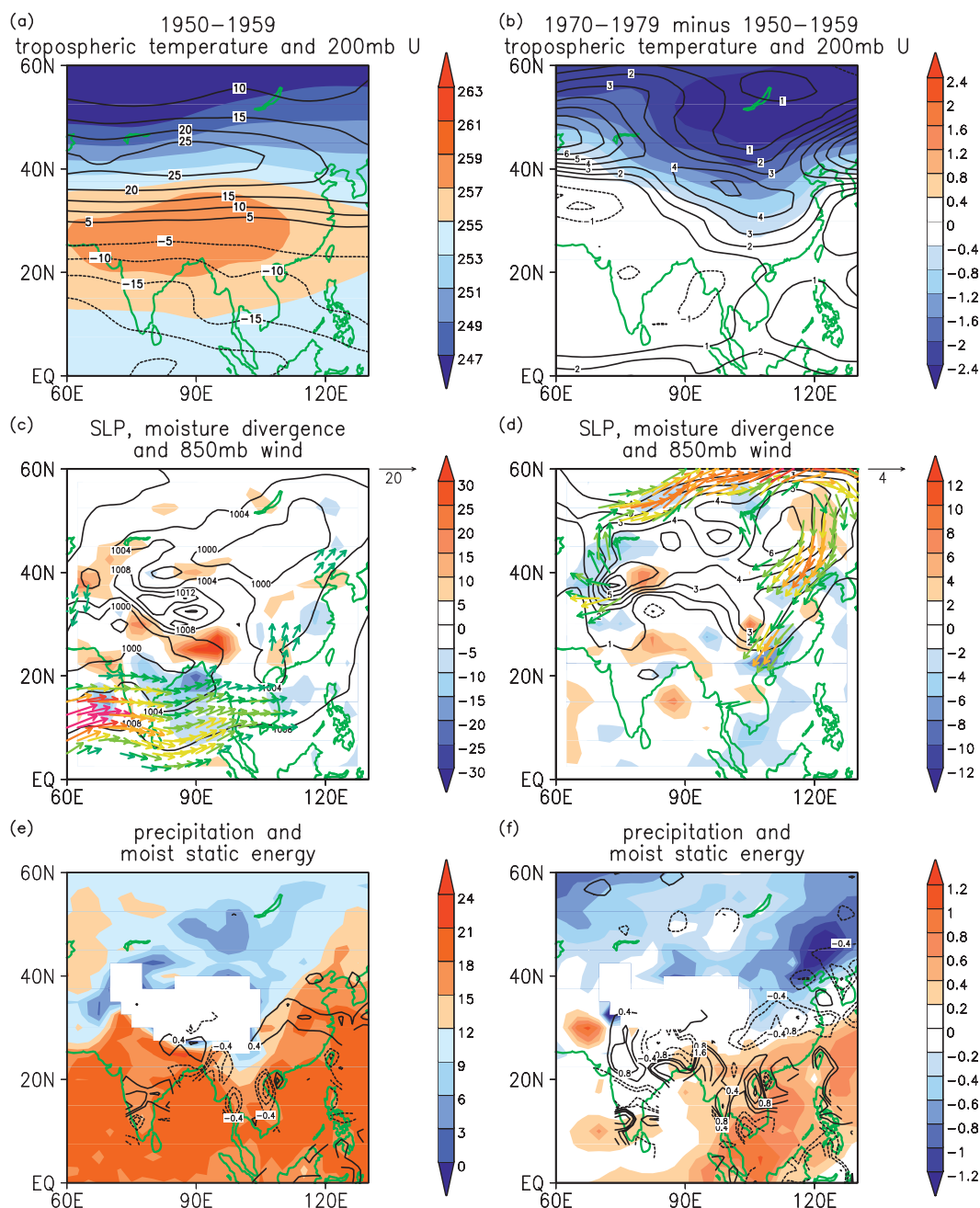


FIG. 8. As in Fig. 6, but for the Asia sector.

is seen where rainfall decreases. SLP increases over extratropical Eurasia and North Africa (Fig. 9g). The models, however, appear to simulate more rainfall over South Asia adjacent to the tropical oceans. This incorrect response may result from a lack of atmosphere–ocean interaction in the prescribed SST runs, as indicated in Zhou et al. (2008). The spatial anomalies in CAM3 CPCA are basically the same, except for warming in the high latitudes of East Asia (Fig. 9, right panels). We

conclude that models forced by the observed SST are able to simulate, in a broad sense, both the temporal and spatial structure of the observed 1960s' shift. The ensemble mean substantially reduces the effects of internal variability, so the shift must originate from the applied forcings. This is consistent with previous modeling studies of the Sahel drought (e.g., Rowell et al. 1995).

We further infer that the forcing causing the 1960s' shift resides in the imposed SST. The ensemble mean of

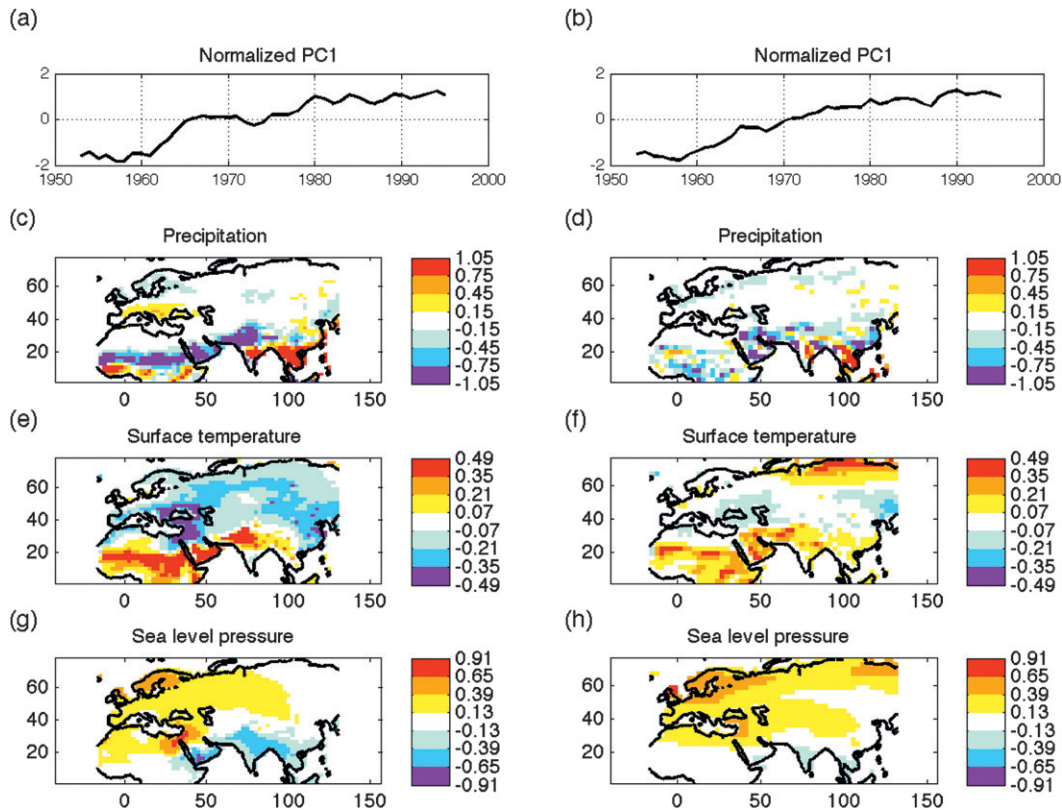


FIG. 9. As in Fig. 5, but for (left) GFDL IAMIP and (right) CAM3 IAMIP.

VAMIP runs from CAM3, in which SST was the only time-varying forcing imposed, was analyzed using the same CPCA procedure. The results are basically the same as that of the CAM3 IAMIP run (figure not shown), suggesting the key role of SST over other forcing agents.

Which ocean basin's SST is responsible for the climate shift? Figure 10 shows regression of global SST onto PC1s from CPCA of both observation and VAMIP simulation. Consistent with previous studies, the regression picks out warming over the tropical oceans and the PDO-like pattern in the Pacific as distinct influences. In addition, however, there is cooling over North Atlantic, especially in the extratropics (Fig. 10, upper panels). Indeed, the signal over North Atlantic is more pronounced in the VAMIP simulation where SST is the only forcing (Fig. 10b). Because we are interested in the abrupt shift in the 1960s, we show in the lower panels of Fig. 10 the same regression, but we restrict the time period from 1950 to 1975 in order to avoid the accelerated warming in late twentieth century and a 1976/77 monsoon transition that originated from the Pacific (Deser and Phillips 2006; Li et al. 2010). The extratropical North Atlantic becomes the region of the largest signal.

6. A simulation of climate impacts to extratropical North Atlantic cooling

Up to now, we have only shown diagnostic relationships between extratropical North Atlantic cooling and the monsoon weakening, with no definitive evidence for causality. In this section, we use an AGCM to demonstrate that extratropical North Atlantic cooling can drive the requisite continental climate changes and monsoon weakening, at least qualitatively.

We use the CAM3 coupled to a fixed-depth slab ocean model that interacts thermodynamically with the atmosphere, but has no representation of ocean dynamics. A monthly varying "Q flux" is applied to the model slab ocean to ensure that the simulated SST resembles the observed SST distribution; the Q flux is derived from surface fluxes extracted from a global climatological fixed-SST simulation. The slab ocean configuration is required, because the thermodynamic ocean-atmosphere interaction is essential to bringing the extratropical influence to the tropics (Chiang and Bitz 2005). Such interaction is also crucial in correctly simulating the South Asian monsoon, as demonstrated above.

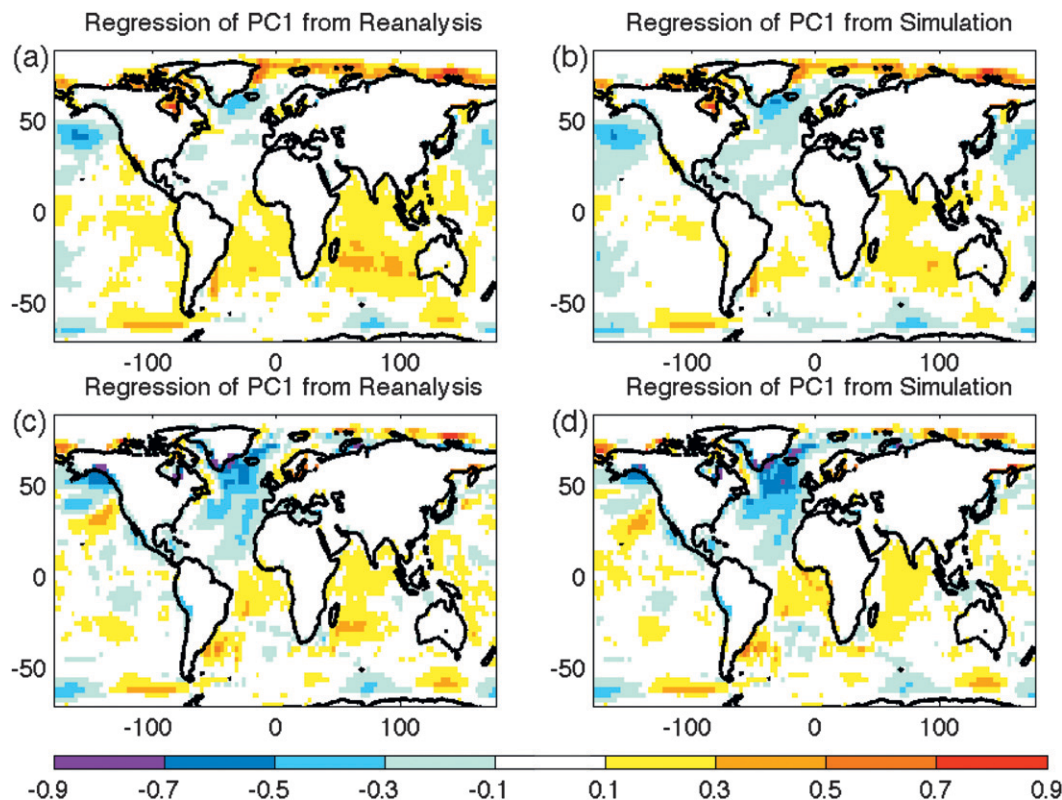


FIG. 10. Regression of June–August global SST onto PC1 from CPCA analysis of (a),(c) NCEP–DAI and (b),(d) CAM3 VAMIP, over the period of (a),(b) 1950–95 and (c),(d) 1950–75. The unit is degrees Celsius per standard deviation of the PC. The SST is produced by merging the monthly mean Hadley Center Sea Ice and SST Dataset and the NOAA weekly optimum interpolation SST as used in the CAM3–VAMIP simulation.

We use the above base configuration to undertake two sets of simulations. In the one set, we replace the slab ocean in the North Atlantic over the 45° – 60° latitude band with climatological monthly varying fixed SSTs, and run this configuration to equilibrium. In the second set, we also apply fixed SSTs over the same North Atlantic region, but uniformly cool the applied SST by a specified amount to represent the extratropical North Atlantic cooling. All simulations are run for 20 yr, and the last 10 yr are taken for climatology.

The June–August anomalies, from a simulation with 4°C cooling applied over the extratropical North Atlantic, are shown in Fig. 11. The climate response is qualitatively similar to the coupled model hosing simulation of Cheng et al. (2007; Fig. 2) and the observed 1960s' climate shift: there is cooling extending over Eurasia and North Africa, rainfall reduces over the Sahel and South Asia, and the monsoon winds over North Africa weaken with an SLP increase in the north. The global SST pattern resembles the one related to AMO in Parker et al. (2007, which will be discussed later) and the interhemispheric pattern in Folland et al. (1986). This

simulation therefore suggests that extratropical North Atlantic cooling can drive the observed Eurasian and North African climate changes associated with the 1960s' shift.

However, we have found in these particular simulations that a North Atlantic extratropical cooling of around 3° – 4°C is required to get an appreciable weakening to the monsoonal rainfall. The observed abrupt shift in extratropical North Atlantic SST in the 1960s is, however, around 1°C , which is significantly smaller than the cooling imposed here (see Fig. 12, e.g.). However, we argue that the apparent mismatch in the climate response's sensitivity to extratropical North Atlantic cooling is to be expected, because the strength of the connection may depend on the magnitude of important feedbacks, of which the two most important appear to be cloud feedbacks and moist convection.

With regards to the former, Kang et al. (2009) showed that the strength of cloud feedbacks determines the magnitude of the tropical response to extratropical cooling. They showed in their aquaplanet simulations with extratropical cooling that switching off cloud

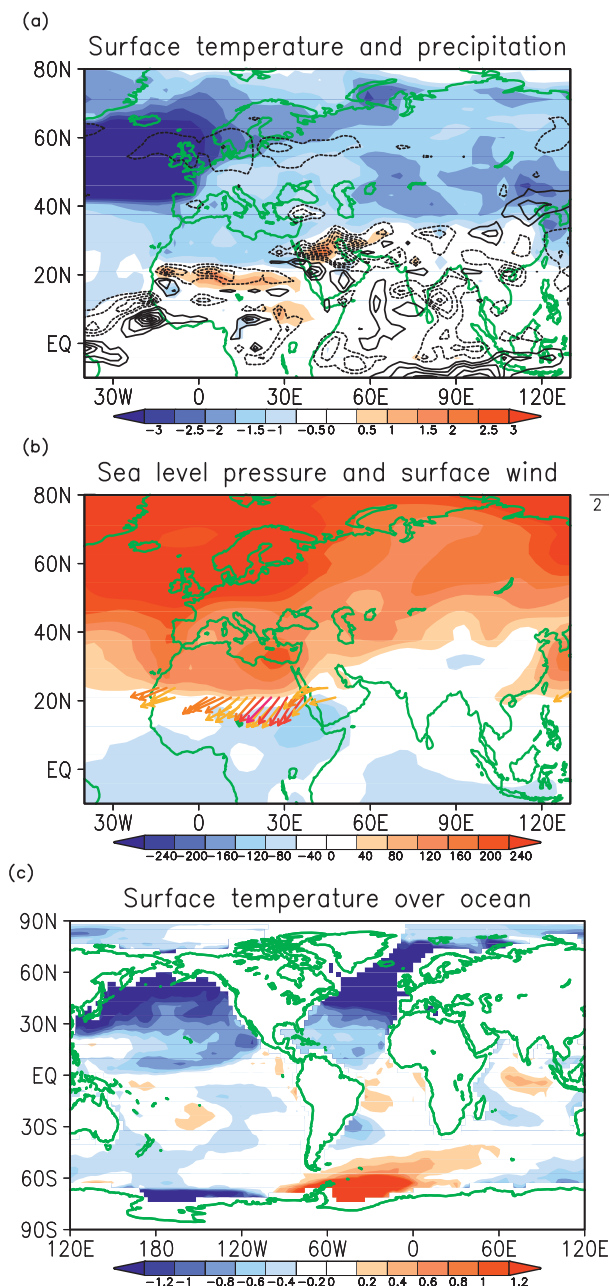


FIG. 11. June–August climate anomalies in an AGCM simulation using CAM3 with 4°C cooling prescribed in between 45° and 60° over North Atlantic. Changes in (a) surface temperature (shaded, K) and rainfall (contour interval 0.5 mm day^{-1} , dashed contours are negative), (b) sea level pressure (shaded, Pa) and surface wind (vectors, m s^{-1} , value under 2 not shown), and (c) surface temperature (K) over global ocean.

feedbacks (through prescribing clouds) effectively reduced the response of Hadley circulation, and thus that of tropical rainfall. Correctly simulating cloud responses to climate changes, however, is notoriously difficult. With regards to convection, the Zhang–McFarlane (ZM)

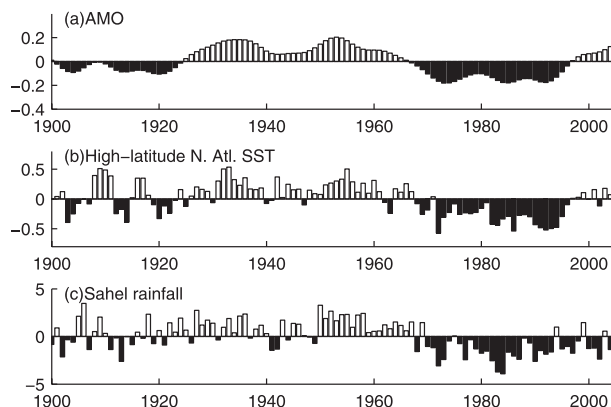


FIG. 12. The time series of the AMO ($^{\circ}\text{C}$), high-latitude North Atlantic SST ($^{\circ}\text{C}$), and the unsmoothed June–October Sahel rainfall (cm month^{-1}). The AMO is computed by Trenberth and Shea (2006) as low-filtered Atlantic SST anomaly north of the equator with global mean SST subtracted. The high-latitude North Atlantic SST time series is computed as SST averaged over 50° – 70°N and 70°W – 0° with global mean SST subtracted, using the HadSST2 dataset (Rayner et al. 2006). The Sahel rainfall index is the same as used in Fig. 1.

convection scheme that is used in the CAM3 model may be a weak link. Chen et al. (2010) compared the ZM scheme with three other convection schemes, and showed that the former is the least able in simulating the East Asian monsoon rainfall climatology. Chen et al. (2010) also argued that because convection in the tropics (subtropics) is more sensitive to surface heating (synoptic-scale wave activity), a convection scheme good at simulating tropical rainfall may turn out to be less effective outside the tropics. Furthermore, Kang et al. (2009) showed that the response of tropical rainfall to extratropical forcing is sensitive to the change in the convection scheme. Taken together, these demonstrate the complexity of the various processes determining model sensitivity to extratropical thermal forcing.

We take this preliminary model result using the CAM3 as encouraging, in that it is able to qualitatively simulate an approximately correct pattern of climate response given extratropical North Atlantic cooling. Quantitatively, however, we argue that it is reasonable to expect mismatches in the magnitude of the response in model simulations (as in the case with our simulations) because of known shortcomings in AGCM model physics. The resolution of this problem requires a detailed investigation of the AGCM response to extratropical North Atlantic cooling, including a focus on understanding teleconnection mechanisms, as well as exploring the role of various feedbacks such as clouds and convection in the teleconnection. This assessment is beyond the scope of this particular study, and we defer this to a later study.

7. Conclusions and discussion

We objectively characterized a summertime climate shift over the monsoonal regions of North Africa and Eurasia that took place during the 1960s, following studies that documented such abrupt shifts over individual regions, especially the rainfall variability over the Sahel (Baines and Folland 2007; Folland et al. 1986). Motivated by paleoproxy studies of abrupt climate changes during the last glacial period that showed coincident changes between extratropical North Atlantic cooling and weakening monsoons (Stager et al. 2011; Gupta et al. 2003; Shanahan et al. 2009; Wang et al. 2001), and modeling studies simulating the impacts of a cooler extratropical North Atlantic that suggested such coherent climate changes (Chiang and Bitz 2005; Cheng et al. 2007), we hypothesized that the abrupt monsoonal changes in the 1960s were similarly coordinated and linked to the extratropical North Atlantic cooling.

Following our hypothesis, we incorporated three monsoon regions—the Sahel, South Asia, and East Asia—as well as the Eurasian and North African landmasses, as part of an objective analysis of climate variations over the latter half of the twentieth century. Surface temperature, sea level pressure (SLP), and precipitation of the second half of twentieth century from both reanalysis and observation were analyzed using combined principal component analysis (CPCA). The leading mode is a large monotonic shift among all fields centered on the 1960s. The associated spatial structure of precipitation anomalies captured the Sahel drought, the “north drought–south flooding” over eastern China, and a less strong drought in South Asia. The temperature pattern showed cooling over Eurasia and North Africa with warming over central Asia and the Sahel. The SLP increased uniformly with maxima in regions corresponding to maximum cooling.

To explore the link between the large-scale climate shift and the observed rainfall reduction over the three monsoon regions, we investigated the regional atmospheric circulation changes, some of which have been characterized by previous studies (Goswami et al. 2006; Biasutti et al. 2009; Nicholson 2009; Li et al. 2010). The results show that the cooling extends throughout the depth of the troposphere from the surface over those regions. Over the Sahel and North China, the cooling shifted or weakened the upper-level jet, which would otherwise induce low-level convergence as seen before the 1960s. In addition, the increase in SLP over land diminished the meridional pressure gradient and weakened the prevailing monsoon. For India, the weakening of monsoon is less prominent, but a southward shift of rainbelt is discernible.

Twentieth-century AGCM simulations that imposed known twentieth-century climate forcings as well as

observed variations in global SST are able to capture the observed climate shift, as indicated by the results of a similar CPCA analysis. This result bolsters the evidence for the existence of this coordinated 1960s' climate shift on the monsoons, and furthermore suggests that it was driven by external climate forcings or SST variations. A further analysis using simulations forced only with observed SST variations suggests that SST variations are the primary cause.

Various lines of evidence point to the extratropical North Atlantic SST cooling as the underlying driver of the climate shift. Regression of SST onto the PC of the leading mode in the CPCA analysis highlights the North Atlantic cooling, particularly in the extratropics. Moreover, a recent study by Thompson et al. (2010) shows convincing evidence of such an abrupt variation in SST in the late 1960s, especially an abrupt cooling over the high-latitude North Atlantic. An AGCM slab ocean simulation with imposed extratropical North Atlantic cooling showed climate impacts qualitatively resembling those seen in the climate shift.

Taken together, our results are consistent with our hypothesis of a coordinated climate shift of the monsoons tied to extratropical North Atlantic cooling. If the results of previous studies are also considered, most significantly Thompson et al. (2010) and Baines and Folland (2007), then the implication is that extratropical North Atlantic cooling may have caused a near-global climate shift during that time, ranging from Northern Hemisphere cooling and subtropical monsoon rainbelt relocation, to a southward shift of ITCZ and shift in the Southern Hemisphere storm track.

We emphasize that our diagnostic study only demonstrates the plausibility of this scenario. More work is needed to advance this hypothesis: in particular, a proper detection and attribution study has yet to be performed, and the mechanisms are yet to be elucidated. In this regard, we are currently undertaking idealized AGCM studies, as exemplified in section 6, to investigate atmospheric teleconnection mechanisms of extratropical North Atlantic cooling to the Northern Hemisphere monsoons, and key feedbacks such as the positive cloud feedbacks that is important in determining the response of the monsoons.

What are the causes of the abrupt cooling in the high-latitude North Atlantic during the late 1960s? In their study on the 1960s' climate shift, Baines and Folland (2007) argued for two potential sources—one being the multidecadal variations in North Atlantic SST known as the AMO, and the other being anthropogenic sulfate aerosol emissions over the Northern Hemisphere, particularly over Europe and North America. We first comment on the latter. It is known that sulfate aerosol

emissions increased significantly in the middle of the twentieth century, in particular over Europe and North America (Smith et al. 2004), and several idealized modeling studies have shown that climate forcing resulting from such emissions cool the Northern Hemisphere and weaken the Northern Hemisphere monsoons, in particular if indirect effects are also included (e.g., Rotstayn and Lohmann 2002). While there is increasing evidence implicating anthropogenic aerosols in tropical rainfall changes, we argue that they are not the main factor in the climate shift of the 1960s. First, our own CPCA analysis of twentieth-century simulations suggests that SST is the main driver (though we point out two caveats: simulations we analyzed only included aerosol direct effects, leaving out indirect effects; and anthropogenic aerosols are likely to have also affected the twentieth century SST evolution, so some effects of aerosols may be folded into the SST). Second, even though the sulfur dioxide emissions ramped up since 1950s and peaked in the 1980s, black carbon emissions over South and East Asia, which have been shown to significantly perturb the monsoons (Smith et al. 2004; Novakov et al. 2003; Ramanathan and Carmichael 2008), increased well into the 1990s. Hence, the total effects of aerosols on the monsoons, if significant, should be largest late in the twentieth century rather than in the 1960s. Third, sulfur dioxide emission increased steadily within the period from 1950 to 1975, unlike the abruptness of the high-latitude North Atlantic cooling in the late 1960s shown in Thompson et al. (2010), or the climate change demonstrated above.

We now discuss the AMO. The AMO is a coherent pattern of multidecadal variability in surface temperature centered in the North Atlantic Ocean, the leading cause of whose change is suggested to be variations in the AMOC (Knight et al. 2005). The phasing of the AMO with regards to our problem is indeed correct: the 1960s was a period when the AMO shifted from a positive to negative phase, resulting in a cooling of the North Atlantic SST; as such, it is more likely to be relevant to the 1960s' shift than sulfate aerosols. However, even the AMO may not be the most appropriate index, depending on the way that it is constructed. It is commonly defined to be the SST anomalies averaged over the entire North Atlantic basin, and then temporally low-pass filtered to extract multidecadal variations [e.g., Sutton and Hodson (2005) used annually averaged SST from 0° – 60° N and 75° – 7.5° W, and the resulting time series was smoothed with a 37-point Henderson filter and then detrended]. As such, this index somewhat lacks a physical basis, but more importantly because of the temporal smoothing it obscures faster and more abrupt changes to the SST (Fig. 12, upper time series). Parker et al. (2007)

uses a different definition of the AMO, the third EOF of low-pass-filtered global SST. The spatial pattern mimics that associated with the Sahel drought in Folland et al. (1986). Their AMO definition therefore is the most appropriate one among others in terms of relating the 1960s' climate shift.

Overall, given that the Sahel drought occurred with equal abruptness to the high-latitude North Atlantic SST drop seen in Thompson et al. (2010) (their Fig. 3b), we argue that those findings are more relevant to the 1960s climate shift discussed here than the smoothed AMO. The middle and bottom time series of Fig. 12 show a high-latitude North Atlantic SST index computed by us, together with the unsmoothed Sahel rainfall variation. The SST time series indeed shows a pronounced shift around the 1960s, coinciding with the start of the Sahel drought.

Finally, what caused the abrupt cooling in high-latitude North Atlantic? One possibility that is discussed in the literature is the Great Salinity Anomaly (GSA) in the late 1960s, when anomalously freshwater and sea ice was introduced into the North Atlantic subpolar gyre. The low-salinity water was advected into the Labrador Sea and reduced the deep-water formation there, leading to a weakening of the thermohaline circulation (Dickson et al. 1988; Zhang and Vallis 2006), and thus cooling the North Atlantic SST. Sea ice feedbacks may also have played a role in amplifying this cooling. Zhang and Vallis (2006) perturbed an ocean general circulation model with freshwater flux analogous to that in the GSA event in the second half of the twentieth century. The modeled SST response compared favorably with observations, with strong cooling south of Greenland. We thus speculate that the GSA in the late 1960s may have cooled extratropical North Atlantic, leading to the climate shifts here discussed.

Acknowledgments. This research is supported by the Office of Science (BER), U.S. Department of Energy (Award DE-FG02-08ER64588 to John Chiang), and National Science Foundation (Award NSF EAR-0909195). Comments by Chris Folland and an anonymous reviewer greatly improved the paper. We thank Inez Fung, Kurt Cuffey, Rob Rhew, Chingyee Chang, Andrew Friedman, Shihyu Lee, and Jian Jin for useful and enlightening discussions. Andrew Friedman assisted in editing the manuscript. We also thank Xinyu Wen and Shaowu Wang for providing north China station rainfall data. The Sahel rainfall index, the CRU temperature and precipitation, and the NCAR SLP data were acquired from the Joint Institute for the Study of the Atmosphere and Ocean (<http://jisao.washington.edu/>). The GFDL simulations were acquired from the IRI/LDEO

Climate Data Library, Columbia University (<http://iridl.ldeo.columbia.edu/>). Some Matlab manuscripts used in our analysis were acquired from Daniel Vimont, University of Wisconsin—Madison (<http://www.aos.wisc.edu/~dvimont/>). The regime shift code was written by Sergei Rodionov and was acquired online at <http://www.beringclimate.noaa.gov/>.

REFERENCES

- Alley, R. B., 2007: Wally was right: Predictive ability of the North Atlantic “conveyor belt” hypothesis for abrupt climate change. *Annu. Rev. Earth Planet. Sci.*, **35**, 241–272.
- Bader, J., and M. Latif, 2003: The impact of decadal-scale Indian Ocean sea surface temperature anomalies on Sahelian rainfall and the North Atlantic Oscillation. *Geophys. Res. Lett.*, **30**, 2169, doi:10.1029/2003GL018426.
- Baines, P., and C. Folland, 2007: Evidence for a rapid global climate shift across the late 1960s. *J. Climate*, **20**, 2721–2744.
- Biasutti, M., and A. Giannini, 2006: Robust Sahel drying in response to late 20th century forcings. *Geophys. Res. Lett.*, **33**, L11706, doi:10.1029/2006GL026067.
- , A. Sobel, and S. Camargo, 2009: The role of the Sahara low in summertime Sahel rainfall variability and change in the CMIP3 models. *J. Climate*, **22**, 5755–5771.
- Bottomley, M., C. K. Folland, J. Hsiung, R. E. Newell, and D. E. Parker, 1990: Global ocean surface temperature atlas “GOSTA.” Meteorological Office and the Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, 20 pp + 313 plates.
- Bretherton, C., C. Smith, and J. Wallace, 1992: An intercomparison of methods for finding coupled patterns in climate data. *J. Climate*, **5**, 541–560.
- Broecker, W. S., D. M. Peteet, and D. Rind, 1985: Does the ocean-atmosphere system have more than one stable mode of operation? *Nature*, **315**, 21–26.
- Brohan, P., J. J. Kennedy, I. Harris, S. F. B. Tett, and P. D. Jones, 2006: Uncertainty estimates in regional and global observed temperature changes: A new dataset from 1850. *J. Geophys. Res.*, **111**, D12106, doi:10.1029/2005JD006548.
- Chen, H., T. Zhou, R. B. Neale, X. Wu, and G. J. Zhang, 2010: Performance of the new NCAR CAM3.5 in East Asian summer monsoon simulations: Sensitivity to modifications of the convection scheme. *J. Climate*, **23**, 3657–3675.
- Cheng, W., C. M. Bitz, and J. C. H. Chiang, 2007: Adjustment of the global climate to an abrupt slowdown of the Atlantic meridional overturning circulation. *Ocean Circulation: Mechanisms and Impacts*, *Geophys. Monogr.*, Vol. 173, Amer. Geophys. Union, 295–314.
- Chiang, J. C. H., and C. M. Bitz, 2005: Influence of high latitude ice cover on the marine intertropical convergence zone. *Climate Dyn.*, **25**, 477–496.
- Collins, W., and Coauthors, 2006: The formulation and atmospheric simulation of the Community Atmosphere Model version 3 (CAM3). *J. Climate*, **19**, 2144–2161.
- Dai, A., I. Fung, and A. DelGenio, 1997: Surface observed global land precipitation variations during 1900–88. *J. Climate*, **10**, 2943–2962.
- Dansgaard, W., and Coauthors, 1993: Evidence for general instability of past climate from a 250-kyr ice-core record. *Nature*, **364**, 218–220.
- D’Arrigo, R., R. Wilson, and J. Li, 2006: Increased Eurasian-tropical temperature amplitude difference in recent centuries: Implications for the Asian monsoon. *Geophys. Res. Lett.*, **33**, L22706, doi:10.1029/2006GL027507.
- Delworth, T. L., and Coauthors, 2006: GFDL’s CM2 global coupled climate models. Part I: Formulation and simulation characteristics. *J. Climate*, **19**, 643–674.
- Deser, C., and A. S. Phillips, 2006: Simulation of the 1976/77 climate transition over the North Pacific: Sensitivity to tropical forcing. *J. Climate*, **19**, 6170–6180.
- Dickson, R., J. Meincke, S. A. Malmberg, and A. J. Lee, 1988: The “Great Salinity Anomaly” in the Northern North Atlantic 1968–1982. *Prog. Oceanogr.*, **20**, 103–151.
- Folland, C. K., T. N. Palmer, and D. E. Parker, 1986: Sahel rainfall and worldwide sea temperatures, 1901–85. *Nature*, **320**, 602–607.
- , J. Owen, M. N. Ward, and A. Colman, 1991: Prediction of seasonal rainfall in the Sahel region using empirical and dynamical methods. *J. Forecasting*, **10**, 21–56.
- GFDL Global Atmospheric Model Development Team, 2004: The new GFDL global atmosphere and land model AM2-LM2: Evaluation with prescribed SST simulations. *J. Climate*, **17**, 4641–4673.
- Giannini, A., R. Saravanan, and P. Chang, 2003: Oceanic forcing of Sahel rainfall on interannual to interdecadal time scales. *Science*, **302**, 1027–1030.
- Goswami, B. N., M. S. Madhusoodanan, C. P. Neema, and D. Sengupta, 2006: A physical mechanism for North Atlantic SST influence on the Indian summer monsoon. *Geophys. Res. Lett.*, **33**, L02706, doi:10.1029/2005GL024803.
- Grist, J., and S. Nicholson, 2001: A study of the dynamic factors influencing the rainfall variability in the West African Sahel. *J. Climate*, **14**, 1337–1359.
- Gupta, A. K., D. M. Anderson, and J. T. Overpeck, 2003: Abrupt changes in the Asian southwest monsoon during the Holocene and their links to the North Atlantic Ocean. *Nature*, **421**, 354–357.
- Haarsma, R. J., F. M. Selten, S. L. Weber, and M. Kluiphuis, 2005: Sahel rainfall variability and response to greenhouse warming. *Geophys. Res. Lett.*, **32**, L17702, doi:10.1029/2005GL023232.
- Held, I., T. L. Delworth, J. Lu, K. L. Findell, and T. R. Knutson, 2005: Simulation of Sahel drought in the 20th and 21st centuries. *Proc. Natl. Acad. Sci. USA*, **102**, 17 891–17 896.
- Hulme, M., 1992: A 1951–80 global land precipitation climatology for the evaluation of general-circulation models. *Climate Dyn.*, **7**, 57–72.
- Hurrell, J. W., J. J. Hack, D. Shea, J. M. Caron, and J. Rosinski, 2008: A new sea surface temperature and sea ice boundary dataset for the Community Atmosphere Model. *J. Climate*, **21**, 5145–5153.
- Janowiak, J. E., 1988: An investigation of interannual rainfall variability in Africa. *J. Climate*, **1**, 240–255.
- Kalnay, E., and Coauthors, 1996: The NCEP/NCAR 40-Year Reanalysis Project. *Bull. Amer. Meteor. Soc.*, **77**, 437–471.
- Kang, S. M., D. M. W. Frierson, and I. M. Held, 2009: The tropical response to extratropical thermal forcing in an idealized GCM: The importance of radiative feedbacks and convective parameterization. *J. Atmos. Sci.*, **66**, 2812–2827.
- Knight, J. R., R. J. Allan, C. K. Folland, M. Vellinga, and M. Mann, 2005: A signature of persistent natural thermohaline circulation cycles in observed climate. *Geophys. Res. Lett.*, **32**, L20708, doi:10.1029/2005GL024233.

- Kucharski, F., and Coauthors, 2009: The CLIVAR C20C project: Skill of simulating Indian monsoon rainfall on interannual to decadal timescales. Does GHG forcing play a role? *Climate Dyn.*, **33**, 615–627.
- Li, H., A. Dai, T. Zhou, and J. Lu, 2010: Responses of East Asian summer monsoon to historical SST and atmospheric forcing during 1950–2000. *Climate Dyn.*, **34**, 501–514.
- Nicholson, S., 2009: On the factors modulating the intensity of the tropical rainbelt over West Africa. *Int. J. Climatol.*, **29**, 673–689.
- Novakov, T., V. Ramanathan, J. E. Hansen, T. W. Kirchstetter, M. Sato, J. E. Sinton, and J. A. Sathaye, 2003: Large historical changes of fossil-fuel black carbon aerosols. *Geophys. Res. Lett.*, **30**, 1324, doi:10.1029/2002GL016345.
- Parker, D., C. Folland, A. Scaife, J. Knight, A. Colman, P. Baines, and B. Dong, 2007: Decadal to multidecadal variability and the climate change background. *J. Geophys. Res.*, **112**, D18115, doi:10.1029/2007JD008411.
- Parthasarathy, B., A. Munot, and D. Kothawale, 1994: All-India monthly and seasonal rainfall series: 1871–1993. *Theor. Appl. Climatol.*, **49**, 217–224.
- Ramanathan, V., and G. Carmichael, 2008: Global and regional climate changes due to black carbon. *Nat. Geosci.*, **1**, 221–227.
- Rayner, N. A., D. E. Parker, E. B. Horton, C. K. Folland, L. V. Alexander, D. P. Rowell, E. C. Kent, and A. Kaplan, 2003: Global analyses of sea surface temperature, sea ice, and night marine air temperature since the late nineteenth century. *J. Geophys. Res.*, **108**, 4407, doi:10.1029/2002JD002670.
- , P. Brohan, D. E. Parker, C. K. Folland, J. J. Kennedy, M. Vanicek, T. J. Ansell, and S. F. B. Tett, 2006: Improved analyses of changes and uncertainties in sea surface temperature measured in situ since the mid-nineteenth century: The HadSST2 dataset. *J. Climate*, **19**, 446–469.
- Ren, B., R. Lu, and Z. Xiao, 2004: A possible linkage in the interdecadal variability of rainfall over North China and the Sahel. *Adv. Atmos. Sci.*, **21**, 699–707.
- Rodionov, S., 2004: A sequential algorithm for testing climate regime shifts. *Geophys. Res. Lett.*, **31**, L09204, doi:10.1029/2004GL019448.
- Rotstayn, L. D., and U. Lohmann, 2002: Tropical rainfall trends and the indirect aerosol effect. *J. Climate*, **15**, 2103–2116.
- Rowell, D. P., C. K. Folland, K. Maskell, and M. N. Ward, 1995: Variability of summer rainfall over tropical North Africa (1906–92): Observations and modelling. *Quart. J. Roy. Meteor. Soc.*, **121**, 669–704.
- Shanahan, T., and Coauthors, 2009: Atlantic forcing of persistent drought in West Africa. *Science*, **324**, 377–380.
- Smith, S. J., R. Andres, E. Conception, and J. Lurz, 2004: Historical sulfur dioxide emissions 1850–2000: Methods and results. Joint Global Change Research Institute, PNNL Rep. 14537, 13 pp.
- Stager, J., D. B. Ryves, B. M. Chase, and F. S. R. Pausata, 2011: Catastrophic drought in the Afro-Asian monsoon region during Heinrich event 1. *Science*, **331**, 1299–1302.
- Sutton, R. T., and D. L. R. Hodson, 2005: Atlantic Ocean forcing of North American and European summer climate. *Science*, **309**, 115–118.
- Thompson, D., J. M. Wallace, J. J. Kennedy, and P. D. Jones, 2010: An abrupt drop in Northern Hemisphere sea surface temperature around 1970. *Nature*, **467**, 444–447.
- Trenberth, K. E., and D. Paolino, 1980: The Northern Hemisphere sea-level pressure data set—Trends, errors and discontinuities. *Mon. Weat. Rev.*, **108**, 855–872.
- , and D. J. Shea, 2006: Atlantic hurricanes and natural variability in 2005. *Geophys. Res. Lett.*, **33**, L12704, doi:10.1029/2006GL026894.
- Vellinga, M., and R. A. Wood, 2002: Global climatic impacts of a collapse of the Atlantic thermohaline circulation. *Climatic Change*, **54**, 251–267.
- Wang, S. W., D. Y. Gong, and Z. H. Chen, 2000: Seasonal precipitation series of eastern China since 1880 and the variability. *Acta Geogr. Sin.*, **67**, 281–293.
- Wang, Y. J., H. Cheng, R. L. Edwards, Z. S. An, J. Y. Wu, C. C. Shen, and J. A. Dorale, 2001: A high-resolution absolute-dated Late Pleistocene monsoon record from Hulu Cave, China. *Science*, **294**, 2345–2348.
- Wilks, D. S., 2006: *Statistical Methods in the Atmospheric Sciences*. Academic Press, 630 pp.
- Xu, Q., 2001: Abrupt change of the mid-summer climate in central east China by the influence of atmospheric pollution. *Atmos. Environ.*, **35**, 5029–5040.
- Yu, R., and T. Zhou, 2007: Seasonality and three-dimensional structure of interdecadal change in the East Asian monsoon. *J. Climate*, **20**, 5344–5355.
- Zhang, R., and T. L. Delworth, 2005: Simulated tropical response to a substantial weakening of the Atlantic thermohaline circulation. *J. Climate*, **18**, 1853–1860.
- , and —, 2006: Impact of Atlantic multidecadal oscillations on India/Sahel rainfall and Atlantic hurricanes. *Geophys. Res. Lett.*, **33**, L17712, doi:10.1029/2006GL026267.
- , and G. K. Vallis, 2006: Impact of great salinity anomalies on the low-frequency variability of the North Atlantic climate. *J. Climate*, **19**, 470–482.
- Zhou, T., R. Yu, H. Li, and B. Wang, 2008: Ocean forcing to changes in global monsoon precipitation over the recent half-century. *J. Climate*, **21**, 3833–3852.